Maximum purity of absolutely PPT states of two qutrits

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We determine the maximum purity of a two-qutrit density operator that remains positive under partial transpose after every global unitary transformation. The exact value is 17/121, and the unique decreasingly ordered maximizing spectrum is $(3,1,1,1,1,1,1,1,1)/11$. The computer-assisted proof uses three necessary linear inequalities from the spectral PPT criterion. Their intersection with the ordered probability simplex is an outer polytope with 49 vertices, all enumerated in exact rational arithmetic. An active-constraint argument proves that the enumeration is complete, including degenerate boundaries. The result gives the optimal purity-only sufficient threshold for a global unitary to produce a negative partial transpose, and the maximal Hilbert–Schmidt distance $4\sqrt{2}/33$ from the maximally mixed state within the absolutely PPT set.

I. INTRODUCTION

A global unitary transformation preserves the spectrum and purity of a quantum state, while it can change the state's entanglement across a fixed bipartition. An absolutely positive-partial-transpose (APPT) state stays PPT along its entire global unitary orbit. The largest purity of such a state therefore determines when purity alone guarantees that some global unitary can produce a negative partial transpose (NPT).

For two qutrits, Hildebrand's spectral criterion expresses APPT as two 3×3 positive-semidefinite conditions [1]. Song and Chen gave an extreme-point criterion and identified the spectrum $(3,1,1,1,1,1,1,1,1)/11$ among the two-eigenvalue extreme points; see Corollary 16, Eq. (10), of their preprint version [2]. Wang, Chen, and Song subsequently classified extreme points with exactly three distinct eigenvalues [3]. These restricted classifications do not by themselves maximize purity over the full spectral domain. Dung and Khoi reported the numerical estimate 0.1405 for the two-qutrit APPT maximum in Table 2 of their preprint [4]. Tran's Theorem 1 gives the upper bound 25/169 in total dimension nine; Remark 1.1 leaves the exact qutrit–qudit maximum open [5].

The closest explicit conjecture is due to Ahiable, Kothakonda, and Winter [6]. Their Theorem 6.3 optimizes purity on an *inscribed* APPT polytope. For local dimensions $m = n = 3$, its value is $\max\{17/121, 1/8\} = 17/121$. Their Conjecture 6.7 asserts equality with the maximum on the full APPT set. We prove precisely its 3×3 case. The attaining spectrum is already known; the contribution here is the matching upper bound on the entire domain and the uniqueness of the equality spectrum. All preprint formula and theorem locators in this paper refer to the versions specified in the bibliography.

II. MODEL AND SHARP BOUND

Fix $\mathcal{H} = \mathbb{C}^3 \otimes \mathbb{C}^3$ and let $\rho \succeq 0$ with $\operatorname{Tr}\rho = 1$. Partial transpose T_B is taken on the second factor in a fixed product basis. The APPT condition is

$$(U\rho U^\dagger)^{T_B} \succeq 0 \qquad \text{for every } U \in U(9). \tag{1}$$

There is no restriction on the global unitary, eigenvalue multiplicities, or rank. Partial transpose is a mathematical test, not an assumed physical channel. The state, its spectrum, and its purity are dimensionless. The problem is static and finite dimensional; no perturbative expansion, dynamical approximation, or spatial boundary condition is imposed.

Write the ordered spectrum as

$$\lambda_1 \geq \cdots \geq \lambda_9 \geq 0, \qquad \sum_{i=1}^{9} \lambda_i = 1, \qquad P(\boldsymbol{\lambda}) = \operatorname{Tr}\rho^2 = \sum_{i=1}^{9} \lambda_i^2. \tag{2}$$

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By Eq. (5) and Corollary 4 of the preprint of Ref. [1], Eq. (1) is equivalent to both of the following matrices being positive semidefinite:

$$L_1(\boldsymbol{\lambda}) = \begin{pmatrix} 2\lambda_9 & \lambda_8 - \lambda_1 & \lambda_6 - \lambda_2 \\ \lambda_8 - \lambda_1 & 2\lambda_7 & \lambda_5 - \lambda_3 \\ \lambda_6 - \lambda_2 & \lambda_5 - \lambda_3 & 2\lambda_4 \end{pmatrix}, \tag{3}$$

$$L_2(\boldsymbol{\lambda}) = \begin{pmatrix} 2\lambda_9 & \lambda_8 - \lambda_1 & \lambda_7 - \lambda_2 \\ \lambda_8 - \lambda_1 & 2\lambda_6 & \lambda_5 - \lambda_3 \\ \lambda_7 - \lambda_2 & \lambda_5 - \lambda_3 & 2\lambda_4 \end{pmatrix}. \tag{4}$$

Let K denote the ordered spectral set satisfying Eqs. (2)–(4), including both semidefinite conditions.

Theorem 1. *For every APPT state on* $\mathbb{C}^3 \otimes \mathbb{C}^3$,

$$\operatorname{Tr} \rho^2 \leq \frac{17}{121}. \tag{5}$$

Equality holds if and only if the ordered spectrum is

$$\boldsymbol{\lambda}_* = \frac{1}{11}(3, 1, 1, 1, 1, 1, 1, 1, 1). \tag{6}$$

The proof below is computer assisted: a finite rational enumeration supplies the vertex certificate, while an analytic argument establishes its completeness. The complete certificate is printed in Appendix A.

III. THREE NECESSARY INEQUALITIES

For $v_1 = (1, 1, 0)^T$, $v_2 = (1, 1, 1)^T$, and $v_3 = (2, 1, 0)^T$, define $g_j(\boldsymbol{\lambda}) = v_j^T L_1(\boldsymbol{\lambda}) v_j/2$. Direct expansion gives

$$g_1(\boldsymbol{\lambda}) = -\lambda_1 + \lambda_7 + \lambda_8 + \lambda_9, \tag{7}$$

$$g_2(\boldsymbol{\lambda}) = -\lambda_1 - \lambda_2 - \lambda_3 + \lambda_4 + \lambda_5 + \lambda_6 + \lambda_7 + \lambda_8 + \lambda_9, \tag{8}$$

$$g_3(\boldsymbol{\lambda}) = -2\lambda_1 + \lambda_7 + 2\lambda_8 + 4\lambda_9. \tag{9}$$

Every $\boldsymbol{\lambda} \in K$ obeys $g_j(\boldsymbol{\lambda}) \geq 0$. In particular, $g_2 \geq 0$ is equivalent to $\lambda_1 + \lambda_2 + \lambda_3 \leq 1/2$.

Define the outer polytope

$$Q = \{\boldsymbol{\lambda} \text{ satisfying Eq. (2)} : g_1(\boldsymbol{\lambda}), g_2(\boldsymbol{\lambda}), g_3(\boldsymbol{\lambda}) \geq 0\}. \tag{10}$$

Then $K \subseteq Q$. Only this inclusion is needed for the upper bound; the three inequalities are not asserted to be a sufficient APPT criterion. In particular, $L_2 \succeq 0$ remains part of the physical domain and will also be checked at the equality spectrum.

The same inequalities can be checked directly from Schmidt coefficients. For $b = \sum_{i=1}^{3} x_i |ii\rangle$, the eigenvalues of $(|b\rangle\langle b|)^{T_B}$ are x_i^2 and $\pm x_i x_j$ for $i < j$. Pairing this spectrum in increasing order with $\boldsymbol{\lambda}$ in decreasing order gives the minimum trace over its unitary orbit [1]. The choices $x = (1, 1, 0), (1, 1, 1), (2, 1, 0)$ respectively give the ordered spectra

$$(-1, 0, 0, 0, 0, 0, 1, 1, 1), \quad (-1, -1, -1, 1, 1, 1, 1, 1, 1), \quad (-2, 0, 0, 0, 0, 0, 1, 2, 4), \tag{11}$$

and hence the three linear forms above. Normalization of b is unnecessary because positive scaling does not change the sign of the test.

IV. COMPLETE EXACT VERTEX CERTIFICATE

A. Completeness of the enumeration

Set $\lambda_{10} = 0$ and introduce gap weights

$$y_k = k(\lambda_k - \lambda_{k+1}), \qquad \lambda_i = \sum_{k=i}^{9} \frac{y_k}{k}. \tag{12}$$

TABLE I. Complete vertex enumeration for Q, grouped by the number s of positive gap weights. These are exact rational results.

s	Number of vertices	Largest purity in the group
1	2	1/8
2	14	17/121
3	21	5/36
4	12	3/22

This is a linear bijection between the full ordered probability simplex and $\{y \in \mathbb{R}^9 : y \geq 0,\ \mathbf{1}^T y = 1\}$. It is the standard representation by the uniform rank-k spectra, also used in Ref. [6]; it imposes no multiplicity assumption. In these coordinates the three extra inequalities are $Ay \geq 0$, where

$$A = \begin{pmatrix} -1 & -\frac{1}{2} & -\frac{1}{3} & -\frac{1}{4} & -\frac{1}{5} & -\frac{1}{6} & 0 & \frac{1}{8} & \frac{2}{9} \\ -1 & -1 & -1 & -\frac{1}{2} & -\frac{1}{5} & 0 & \frac{1}{7} & \frac{1}{4} & \frac{1}{3} \\ -2 & -1 & -\frac{2}{3} & -\frac{1}{2} & -\frac{2}{5} & -\frac{1}{3} & -\frac{1}{7} & \frac{1}{8} & \frac{5}{9} \end{pmatrix}. \tag{13}$$

Lemma 1. *Each vertex of $\{y \geq 0 : \mathbf{1}^T y = 1, Ay \geq 0\}$ has at most four positive coordinates. If its positive support is S, with $s = |S|$, some $s-1$ active rows of A, together with the trace row, form a nonsingular $s \times s$ system on S whose solution is the vertex.*

Proof. Within the simplex face with positive support S, the space of directions d supported on S and satisfying $\mathbf{1}^T d = 0$ has dimension $s-1$. If the active rows of A do not eliminate this space, there is a nonzero direction d that also annihilates all active rows. For sufficiently small $\epsilon > 0$, both $y + \epsilon d$ and $y - \epsilon d$ preserve the positive coordinates and all strictly slack inequalities. They are distinct feasible points with midpoint y, contradicting vertexhood. Thus $s-1$ independent active restrictions exist. Since A has only three rows, $s \leq 4$; choosing such restrictions gives the stated nonsingular system. □

Consequently, every vertex is found by considering $s = 1, \ldots, 4$, every support $S \subseteq \{1, \ldots, 9\}$ of size s, and every subset J of $s-1$ rows of A. For each nonsingular matrix

$$M_{S,J} = \begin{pmatrix} \mathbf{1}_S^T \\ A_{J,S} \end{pmatrix}, \qquad M_{S,J} y_S = (1, 0, \ldots, 0)^T, \tag{14}$$

solve exactly, set $y_{S^c} = 0$, and check all inequalities $y \geq 0$ and $Ay \geq 0$. Each feasible solution is a vertex: the selected independent equalities and the zero coordinates determine it uniquely. Solutions with zero weights are retained and deduplicated, so degeneracies are included. There are exactly

$$\sum_{s=1}^{4} \binom{9}{s} \binom{3}{s-1} = 9 + 108 + 252 + 126 = 495 \tag{15}$$

systems to consider. Singular systems may be discarded because Lemma 1 supplies a nonsingular basis for every vertex. The bound on positive gaps concerns vertices of the outer polytope and is a consequence of its three cuts, not a restriction on arbitrary APPT states.

B. Rational evaluation and cross-checks

Exact enumeration gives 72 singular systems and 423 nonsingular systems. There are 63 feasible bases, which yield 49 distinct vertices after removing duplicates. Table I summarizes the results by the actual positive support size. Every vertex and its exact purity are listed in Appendix A. Purity can be evaluated directly from the listed weights using

$$P(y) = \sum_{k,l=1}^{9} \frac{y_k y_l}{\max(k,l)}, \tag{16}$$

which follows by substituting Eq. (12) into $\sum_i \lambda_i^2$.

The unique vertex with purity 17/121 has $y_1 = 2/11$, $y_9 = 9/11$, and all other gap weights zero. Every other vertex has purity at most 5/36, with a strict gap

$$\frac{17}{121} - \frac{5}{36} = \frac{7}{4356} > 0. \tag{17}$$

For a separate representation of the finite calculation, one may work directly in the nine spectrum coordinates. There are twelve homogeneous inequalities: eight ordering inequalities, $\lambda_9 \geq 0$, and the three $g_j \geq 0$. Enumerating all $\binom{12}{8} = 495$ choices of eight active inequalities, together with the trace row, and solving by rational Gauss–Jordan elimination reproduces all 49 vertices. The supplied standard-library Python program implements this calculation and checks the complete distributed table only after generating its vertex set. A separate exact incremental clipping calculation also reproduces the same set; adding g_1, g_2, g_3 in that order gives 15, 31, and 49 vertices and maxima 25/169, 1/7, and 17/121. The first step recovers the dimension-nine relaxation underlying Ref. [5].

All decisions in the certificate use rational arithmetic, with no feasibility tolerance. Floating-point optimization is not part of the proof. Agreement between implementations is a cross-check; completeness rests on the active-constraint argument above. Regeneration instructions and the exact data accompany the manuscript.

C. Research tools and verification

Generative AI (OpenAI Codex) was used substantively for literature exploration, selection of test vectors, development of the proof, preparation and execution of exact-arithmetic code, internal checks, and manuscript drafting. The computational checks consisted of complete rational enumerations in different representations and a separate rational clipping implementation. These checks do not constitute human peer review. This local draft still requires the named author's scientific review and final approval; no completed human verification is asserted.

V. PROOF OF THE SHARP BOUND AND ITS CONSEQUENCES

Proof of Theorem 1. The compact polytope Q is the convex hull of its vertices. Since $P(\boldsymbol{\lambda}) = \|\boldsymbol{\lambda}\|_2^2$ is convex, the exact vertex bound implies $P \leq 17/121$ on all of Q, hence on $K \subseteq Q$. Strict convexity rules out a nonvertex maximizer: a nontrivial convex combination of distinct feasible points has purity strictly below the corresponding convex combination of their purities. The only possible ordered equality spectrum is therefore the unique maximizing vertex $\boldsymbol{\lambda}_*$.

At this spectrum, both complete semidefinite constraints become

$$L_1(\boldsymbol{\lambda}_*) = L_2(\boldsymbol{\lambda}_*) = \frac{2}{11}\begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \succeq 0. \tag{18}$$

Their eigenvalues are $0, 2/11, 4/11$. Thus $\boldsymbol{\lambda}_* \in K$, and $P(\boldsymbol{\lambda}_*) = (9+8)/121 = 17/121$, proving attainment and uniqueness. □

Attainment also has a direct full-orbit check. Every state with spectrum $\boldsymbol{\lambda}_*$ has the form

$$\rho = \frac{I_9 + 2|\psi\rangle\langle\psi|}{11} \tag{19}$$

for a normalized $|\psi\rangle$. If its Schmidt coefficients are x_i, the potentially smallest eigenvalues of ρ^{T_B} are $(1 - 2x_i x_j)/11$. They are nonnegative because $2x_i x_j \leq x_i^2 + x_j^2 \leq 1$.

For any two-qutrit state, Theorem 1 implies

$$\operatorname{Tr}\rho^2 > \frac{17}{121} \quad \Longrightarrow \quad \exists U \in U(9) : (U\rho U^\dagger)^{T_B} \not\succeq 0. \tag{20}$$

This purity-only sufficient threshold cannot be lowered, since the APPT orbit in Eq. (19) reaches it. The implication is not a characterization of states at lower purity: the theorem does not assert that every state below the threshold is APPT. It also does not assert distillability of the resulting NPT state or identify APPT with absolute separability.

Finally, for the Hilbert–Schmidt norm $\|X\|_{\mathrm{HS}} = (\mathrm{Tr}\, X^\dagger X)^{1/2}$,

$$\max_{\rho\in \mathrm{APPT}_{3,3}} \left\| \rho - \frac{I_9}{9} \right\|_{\mathrm{HS}}^2 = \frac{17}{121} - \frac{1}{9} = \frac{32}{1089}. \tag{21}$$

The maximum distance is therefore $4\sqrt{2}/33$, attained precisely by the unitary orbit in Eq. (19).

VI. DISCUSSION

Three necessary linear tests suffice to enclose the full two-qutrit APPT spectral domain in a polytope with the correct largest purity. The exact upper bound improves 25/169 by 152/20449 and matches the previously known attaining spectrum. This establishes the 3×3 instance of the maximum-purity conjecture of Ref. [6], including its maximizing spectrum. No conclusion for general $3 \times n$ or other local dimensions follows from this nine-variable certificate.

The argument avoids a classification of all APPT extreme points: it classifies only the vertices of a deliberately larger set. Its limitation is equally specific. The finite rational table and the analytic completeness argument together give a computer-assisted proof, rather than a floating-point search or a proof-assistant formalization. An analytic compression of the certificate could simplify the presentation, but is not assumed here.

DATA AVAILABILITY

All exact vertex data and a Python standard-library regeneration program are included with this local manuscript in the accompanying reproducibility materials. The full vertex table is also printed below. No experimental or third-party numerical dataset is used, and no public repository deposit is claimed.

Appendix A: Complete rational vertex certificate

Table II gives all vertices of Q. An entry $k : a/b$ means $y_k = a/b$; every unlisted weight is zero. The ordered spectrum is recovered from Eq. (12). Thus the table, Eq. (13), and Eq. (16) give a finite exact certificate whose entries can be checked without numerical tolerances. Table numbering fixes one ordering of tied purities and has no mathematical significance.

TABLE II: All 49 vertices of the three-cut outer polytope, in gap weights.

No.	Positive gap weights $k : y_k$	P
1	$1 : 2/11,\ 9 : 9/11$	17/121
2	$1 : 1/12,\ 2 : 1/6,\ 9 : 3/4$	5/36
3	$1 : 1/22,\ 2 : 2/11,\ 8 : 4/11,\ 9 : 9/22$	3/22
4	$2 : 7/32,\ 8 : 1/2,\ 9 : 9/32$	279/2048
5	$4 : 2/9,\ 5 : 5/18,\ 9 : 1/2$	11/81
6	$5 : 10/19,\ 9 : 9/19$	49/361
7	$4 : 4/15,\ 5 : 1/6,\ 8 : 4/15,\ 9 : 3/10$	61/450
8	$4 : 14/39,\ 8 : 16/39,\ 9 : 3/13$	206/1521
9	$4 : 1/3,\ 7 : 7/24,\ 9 : 3/8$	13/96
10	$2 : 1/4,\ 9 : 3/4$	13/96
11	$1 : 1/8,\ 3 : 1/8,\ 9 : 3/4$	13/96
12	$4 : 1/3,\ 6 : 1/10,\ 8 : 4/15,\ 9 : 3/10$	27/200
13	$2 : 1/5,\ 7 : 7/20,\ 9 : 9/20$	27/200
14	$1 : 3/20,\ 8 : 2/5,\ 9 : 9/20$	27/200
15	$5 : 15/32,\ 8 : 1/4,\ 9 : 9/32$	69/512
16	$2 : 1/5,\ 6 : 3/25,\ 8 : 8/25,\ 9 : 9/25$	84/625
17	$3 : 1/12,\ 5 : 5/12,\ 9 : 1/2$	29/216
18	$4 : 1/3,\ 6 : 1/6,\ 9 : 1/2$	29/216
19	$3 : 1/10,\ 5 : 1/3,\ 8 : 4/15,\ 9 : 3/10$	2/15
20	$4 : 2/5,\ 9 : 3/5$	2/15
21	$2 : 1/5,\ 6 : 1/5,\ 9 : 3/5$	2/15
22	$5 : 5/13,\ 7 : 7/26,\ 9 : 9/26$	45/338
23	$1 : 1/26,\ 4 : 4/13,\ 8 : 4/13,\ 9 : 9/26$	45/338
24	$2 : 2/13,\ 5 : 5/26,\ 8 : 4/13,\ 9 : 9/26$	45/338
25	$2 : 1/8,\ 5 : 5/16,\ 9 : 9/16$	17/128

Continued on next page

Table II continued

No.	Positive gap weights $k : y_k$	P
26	$1 : 1/14,\ 4 : 2/7,\ 9 : 9/14$	13/98
27	$1 : 1/11,\ 3 : 3/22,\ 8 : 4/11,\ 9 : 9/22$	16/121
28	$3 : 1/8,\ 5 : 5/24,\ 7 : 7/24,\ 9 : 3/8$	19/144
29	$3 : 7/34,\ 8 : 12/17,\ 9 : 3/34$	38/289
30	$7 : 35/44,\ 9 : 9/44$	127/968
31	$3 : 2/11,\ 7 : 21/44,\ 9 : 15/44$	127/968
32	$3 : 1/6,\ 6 : 1/6,\ 7 : 7/24,\ 9 : 3/8$	113/864
33	$3 : 1/6,\ 6 : 4/15,\ 8 : 4/15,\ 9 : 3/10$	88/675
34	$3 : 3/19,\ 8 : 16/19$	47/361
35	$6 : 3/7,\ 7 : 1/4,\ 9 : 9/28$	51/392
36	$4 : 1/5,\ 8 : 4/5$	13/100
37	$1 : 1/20,\ 3 : 3/20,\ 7 : 7/20,\ 9 : 9/20$	13/100
38	$6 : 18/35,\ 8 : 8/35,\ 9 : 9/35$	159/1225
39	$2 : 1/9,\ 8 : 8/9$	7/54
40	$3 : 1/6,\ 6 : 1/3,\ 9 : 1/2$	7/54
41	$1 : 1/9,\ 7 : 7/18,\ 9 : 1/2$	7/54
42	$5 : 5/21,\ 8 : 16/21$	19/147
43	$6 : 4/7,\ 9 : 3/7$	19/147
44	$7 : 7/15,\ 8 : 8/15$	29/225
45	$6 : 3/11,\ 8 : 8/11$	31/242
46	$1 : 1/17,\ 8 : 16/17$	37/289
47	$8 : 1$	1/8
48	$3 : 1/4,\ 9 : 3/4$	1/8
49	$9 : 1$	1/9

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